

- 6 Lactate dehydrogenase (LDH) is an enzyme used in some blood tests to help diagnose liver disease.

In liver disease, an enzyme known as ALT can leak out from damaged liver tissue and enter the blood.

The presence of ALT can be confirmed by testing a blood sample at room temperature:

- ALT reacts with substances that are added to the blood sample to produce substrates for the reaction catalysed by LDH
- the rate of reaction of LDH can be used to estimate the concentration of ALT in the blood.

Scientists investigated the activity of LDH extracted from the bacterium *Lactobacillus delbrueckii* at different pH values and at different temperatures.

Figure 6.1 shows the effect of pH on the activity of LDH from *L. delbrueckii*.

Figure 6.2 shows the effect of temperature on the activity of LDH from *L. delbrueckii*.

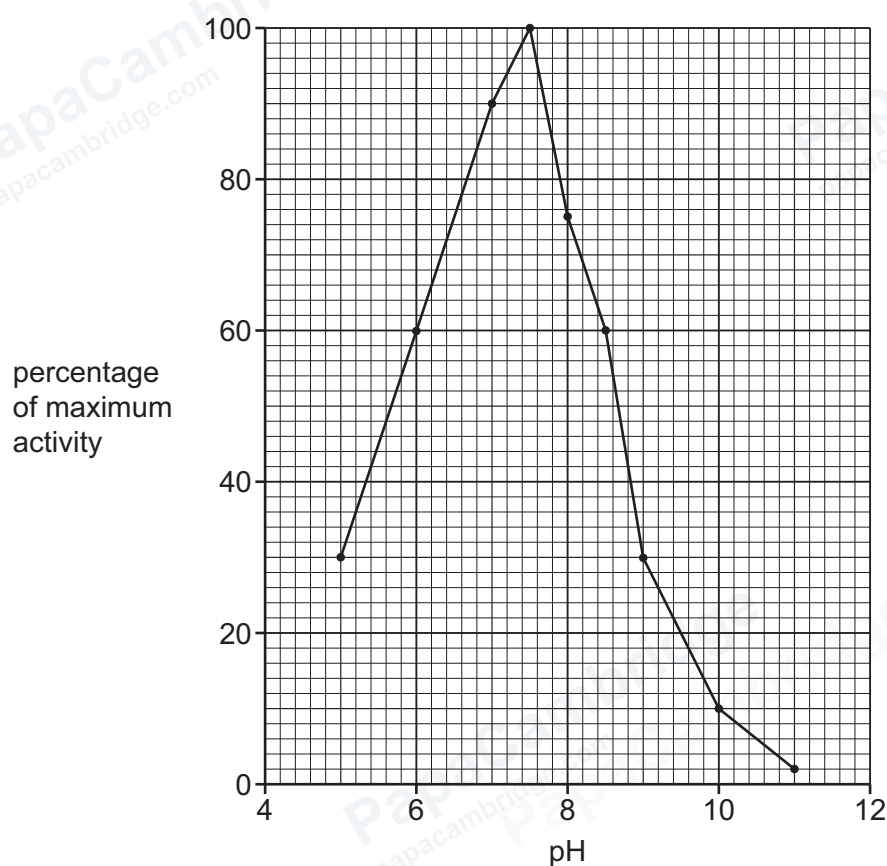


Figure 6.1

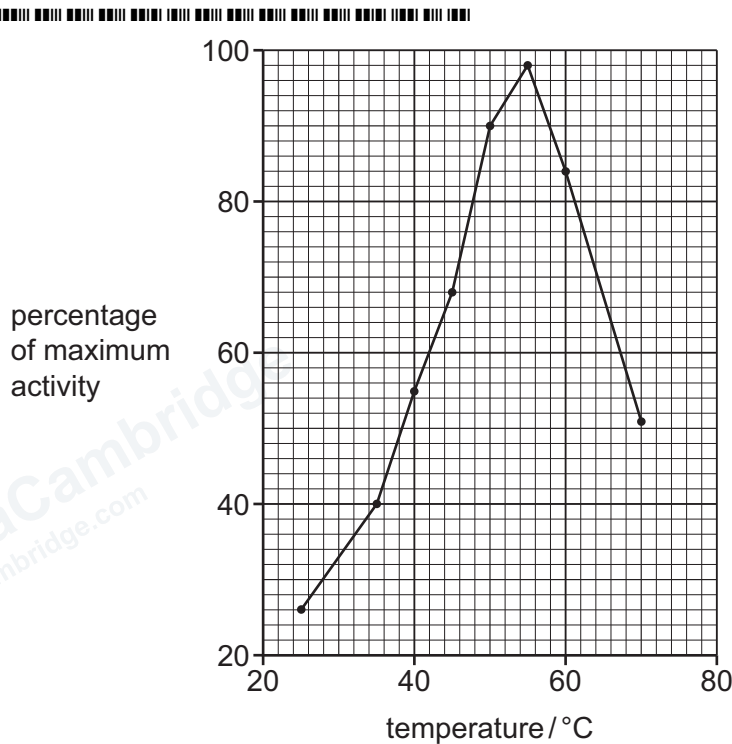


Figure 6.2

- (a) With reference to Figure 6.1 and Figure 6.2, state the optimum pH and the optimum temperature for LDH.

optimum pH

optimum temperature

[2]

- (b) With reference to Figure 6.1, explain how pH can affect the activity of an enzyme such as LDH.

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- (c) Suggest why the recommended temperature for performing the test on the blood sample is not the same as the optimum temperature for LDH activity.

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[Total: 6]